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Foundational Concepts of Analytics:

1. Business Analytics:

- Drive business decision using data to better plan for the future. Using the information to make smart moves for the company's growth.

Example:

- If a hospital uses data to decide where to build a new wing based on where most patients live.

Source: Brightlearn.

2. Data Analytics:

- Processing raw numbers, cleaning and inspecting data to find useful information.

Example:

- pulling a report of all patient admissions and look for errors or missing ID numbers to make sure the list is clean.

Source: Brightlearn.

3. Business Intelligence (BI)

- Taking raw data and turning it into easy-to-read reports so managers can monitor daily operations.

Example:

- Updating the daily slide that the doctors use in their meetings to see how many beds are currently available in each ward.

Source: McKinsey & Company

4. Data-Driven Decision Making

- Making decisions using data as evidence.

Example:

- Instead of the Head Nurse just guessing how many staff are needed, the hospital uses Data-Driven Decision Making by looking at the actual patient count from the last six months to create the new schedule.

Source: Investopedia

5. Big Data

- Big tables and data in different multiple tables, stored in a Data warehouse.

Example:

- In a hospital, years of patient history all being stored in the system at once.

Source: Bright Learn.

6. Descriptive Analytics

- looking at past data to give a summary of what has already taken place.

Example:

- A report showing that the hospital treated 500 patients last week is descriptive analytics, tells us the history without explaining why it happened.

Source: Gartner.

7. Diagnostic Analytics

- focuses on finding the root cause of a specific trend or outlier in the data.

Example:

- If a hospital notices a sudden spike in post-surgery infections, diagnostic analytics would be used to drill down into the data to see if the infections are linked to a specific operating room, a certain surgical team or shared medical supplies.

Source: Gartner

8. Predictive Analytics

- Using historical patterns to make a "best guess" about what might happen next.

Example:

- By looking at data from previous winters, the hospital uses Predictive Analytics to estimate that they will likely see an increase in flu patients next month.

Source: Statista

9. Prescriptive Analytics

- Uses data to answer "what should we do?" by suggesting specific action plans based on the results of descriptive and predictive models.

Example:

- An AI system uses prescriptive analytics to analyze a patient's worsening vitals and automatically suggests the exact dosage of medication the nurse should prepare.

Source: Harvard

10. Data Literacy

- The ability to read, understand and talk about data comfortably. It's a ~~soft~~ skill that helps you ask the right questions when you see a chart or a report.

Example:

- If a doctor looks at a graph of patient recovery times and understands exactly what the average line means.

Source: Cisco Networking Academy

11. Structured Data

- Data that fits perfectly into a table, organized into specific columns.

Example:

- When capturing patient details into the hospital system their age, gender and date of admissions are all structured data because they go into fixed boxes.

Source: Investopedia

12. Unstructured Data

- Data that does not fit into a simple spreadsheet, harder to analyze

Example:

- A doctor's handwritten clinical notes or patient x-ray images.

Source: IBM Doctor Management

13. Data Warehouse

- It is a central data storage, houses multiple databases.

Example:

- A hospital Data Warehouse allows the board to see a single report that combines how much the pharmacy spent on medicine with how many patients were actually treated in the emergency department.

Source: Brightleaf.

14. Data Mining

- This is the process of digging through huge amounts of data to find hidden patterns or trends that are not obvious at first glance.

Example:

Looking at years of patient records to discover certain illnesses that always spike after a change in the local weather.

Source: Statista.

15. Data Visualization

- Using charts and graphs to make information / data easier to understand.

Example:

- Instead of receiving a list of daily admissions, I can create a Bar Graph that shows which day of the week the Hospital is busiest.

Source: Harvard.

16. Dashboard

- A dashboard is a live screen that displays the most important charts and metrics in real-time.

Example:

- Using a dashboard to see a live count of how many patients are currently waiting in the lobby versus doctors on duty.

Source: McKinsey & Company

17. Metric

- Any number that you can track, a way to measure a specific activity even if that activity is not important.

Example:

- The number of lightbulbs changed in the hospital corridors.

Source: Inverseaction.

18. Key Performance Indicator (KPI)

- KPI's tell us if we are reaching the main goals.

Example :

- Patient Mortality Rate, critical number that shows if the hospital is successfully saving lives.

Source : Gartner

19. Benchmark

- Standard number that is used to compare performance against to see if the performance is well or falling behind.

Example :

- If the national Benchmark for emergency waiting times is 4 hours and our hospital takes 6 hours, the benchmark tells us we need to improve our efficiency.

Source : Statista

20. Segmentation

- Breaking a big group into smaller slices based on things they have in common.

Example :

- Using segmentation to group patients by their age to see which group needs more resources.

Source : McKinsey & Company

21. Conversion Rate

- The percentage of people who take a specific action after visiting you.

Example :

- If 100 people walk into the hospital asking about flu shots, but only 20 stay to get the shot, the conversion Rate is 20%.

Source : Investopedia

22. Customer Acquisition Cost (CAC)

- Total amount of money spent to get one new person through the door includes all the marketing and advertising costs.

Example :

- If the hospital spends \$1000 on posters to promote a new health check-up and 10 new patients come in the CAC is \$100 per patient.

Source : Harvard.

23. Customer Lifetime Value (CLV/LTV)

- An estimate of the total money a person will spend with your business for as long as they remain a customer.

Example

- A patient with a chronic condition might visit the hospital every month for 10 years. Their CLV is the total value of those years of appointments and medications combined.

Source : McKinsey & Company.

24. Churn Rate

- Percentage of people who stop using your service over a certain period.

Example :

- If we have 100 regular patients at the clinic this month but 5 of them decide to start going to a different clinic next month, The churn Rate is 5%.

Source : Statista

25. Customer Retention Rate

- Measures how many people stay and keep coming back over time.

Example

- Moms who had their first baby at the hospital and choosing to come back to the hospital for their second baby.

Source : Gartner

26. Net Promoter Score (NPS)

- This is a score that shows how likely people are to recommend your service to their friends or family.

Example :

- Asking a patient to rate our services after discharged from 1 to 10, the average that answer give us our NPS.

Source : Forbes Advisor.

27. Average Order Value (AOV)

- Average amount of money spent everytime a person makes a transaction.

Example:

- If one patient pays R500 for a consultation and another pays R1500 because they also needed blood tests; The AOV for these two patients/visits is R1000.

Source: Investopedia

28. Sales Funnel

- The journey a person takes from first hearing about you to finally paying for a service.

Example:

- Hospital Sales Funnel starts with people seeing our health awareness ad to visiting our website and finally booking an appointment.

Source: McKinsey & Company.

29. Marketing Campaign

- Planned set of activities designed to promote a specific goal.

Example:

- Hospital running a Diabetes Awareness Month in November with free sugar testing and educational posters.

Source: Invisioption.

30. Post Campaign Analysis

- This is the review stage where you look at the data after a campaign is finished to see if it was successful or not.

Example:

- After the diabetes drive ends, a post campaign analysis is done to see if the number of people coming in for check ups actually increased compared to a normal month.

Source: McKinsey & Company

31. Incremental Revenue

- It is the profit you would not have made if you had not run the campaign.

Example:

- If the hospital makes £10,000 from flu shots, but makes £15,000 after sending out a reminder SMS the Incremental Revenue is £5,000.

Source: Harvard.

32. Return on Investment (ROI)

- Measures how much profit you made compared to how much you spent.

Example:

- If we spend £2,000 on a health ad and brings in £6,000 in new patient fees, the ROI is high because the profit is much more than the cost.

Source: Statista.

33. Return on Ad Spend (ROAS)

- Helps you see which specific ads are the most effective at bringing in revenue.

Example:

- If the hospital spends £500 on Facebook Ads and gets £2,000 in bookings from them the ROAS is 4:1 (£4 back for every £1 spent).

Source: Forber Advisor

34. Click-Through Rate (CTR)

- The percentage of people who see a digital link and actually click on it.

Example:

- If ~~200~~ 1000 people see our "Book an eye test" button on the website, but only 50 people click it, the CTR is 5%.

Source: IBM Data Science.

35. Bounce Rate

- The percentage of visitors who click on your website but leave immediately without clicking on anything else.

Example:

- If someone clicks on an ad for Emergency Services but leaves the page because it's too slow to load.

Source: Gardner

36. A/B Testing

- Comparing two versions of something to see which one performs better.

Example:

Sending a blue email to half our patients and a green to email to the other half to see which colour gets more people to book an appointment.

Source: Haward.

37. Lead Generation

- Process of identifying and capturing the contact details of people who are interested in your services so you can follow up later.

Example:

- When a visitor fills out a contact me form on the hospital website to ask about maternity packages.

Source: Investopedia

38. Marketing Attribution

- Process of figuring out exactly which marketing channel deserves credit for patient making a booking.

Example:

- If a patient saw a billboard, then got an SMS and finally booked a check up, help us decide which of these things actually convinced them to come in.

Source: McKinsey & Company.